

Proof of Lemma 4.2 Using the optimality condition, we have:

$$\nabla f(x_k) + J_F(x_k)^T (\rho \operatorname{sign}(l_F(x_{k+1}; x_k)) \circ |l_F(x_{k+1}; x_k)|^{q-1}) + \beta_{k+1} \Delta x_{k+1} = 0.$$

Exploiting definition of $\mathcal{P}_\rho^q$ and properties of the derivative, it follows that:

$$\begin{aligned} & \nabla \mathcal{P}_\rho^q(x_{k+1}) \\ &= \nabla f(x_{k+1}) + J_F(x_{k+1})^T (\rho \operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1}) \\ &= \nabla f(x_{k+1}) - \nabla f(x_k) - \beta_{k+1} \Delta x_{k+1} \\ & \quad + \rho (J_F(x_{k+1}) - J_F(x_k))^T \operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1} \\ & \quad + \rho J_F(x_k)^T (\operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1} - \operatorname{sign}(l_F(x_{k+1}; x_k)) \circ |l_F(x_{k+1}; x_k)|^{q-1}). \end{aligned}$$

It then follows by applying the norm:

$$\begin{aligned} & \|\nabla \mathcal{P}_\rho^q(x_{k+1})\| \\ & \leq \|\nabla f(x_{k+1}) - \nabla f(x_k)\| + \beta_{k+1} \|\Delta x_{k+1}\| \\ & \quad + \rho \|\operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1}\| \|J_F(x_{k+1}) - J_F(x_k)\| \\ & \quad + \rho \|J_F(x_k)\| \|\operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1} - \operatorname{sign}(l_F(x_{k+1}; x_k)) \circ |l_F(x_{k+1}; x_k)|^{q-1}\| \\ & \stackrel{(4)}{\leq} \|\nabla f(x_{k+1}) - \nabla f(x_k)\| + \beta_{k+1} \|\Delta x_{k+1}\| \\ & \quad + \rho \|\operatorname{sign}(F(x_{k+1})) \circ |F(x_{k+1})|^{q-1}\| \|J_F(x_{k+1}) - J_F(x_k)\| \\ & \quad + \rho \|J_F(x_k)\| \left(3 \times m^{\frac{2-q}{2}} \|F(x_{k+1}) - l_F(x_{k+1}; x_k)\|^{q-1} \right) \\ & = \|\nabla f(x_{k+1}) - \nabla f(x_k)\| + \beta_{k+1} \|\Delta x_{k+1}\| \\ & \quad + \rho \left(\sum_{i=1}^m (|f_i(x_{k+1})|^2)^{q-1} \right)^{\frac{1}{2}} \|J_F(x_{k+1}) - J_F(x_k)\| \\ & \quad + \rho \|J_F(x_k)\| \left(3 \times m^{\frac{2-q}{2}} \|F(x_{k+1}) - l_F(x_{k+1}; x_k)\|^{q-1} \right) \\ & \leq \|\nabla f(x_{k+1}) - \nabla f(x_k)\| + \beta_{k+1} \|\Delta x_{k+1}\| \\ & \quad + \rho \left(\sum_{i=1}^m (|f_i(x_{k+1})|^2) \right)^{\frac{q-1}{2}} \times \left(\sum_{i=1}^m 1 \right)^{\frac{2-q}{2}} \|J_F(x_{k+1}) - J_F(x_k)\| \\ & \quad + \rho \|J_F(x_k)\| \left(3 \times m^{\frac{2-q}{2}} \|F(x_{k+1}) - l_F(x_{k+1}; x_k)\|^{q-1} \right) \\ & = \|\nabla f(x_{k+1}) - \nabla f(x_k)\| + \beta_{k+1} \|\Delta x_{k+1}\| \\ & \quad + m^{\frac{2-q}{2}} \rho \|F(x_{k+1})\|^{q-1} \|J_F(x_{k+1}) - J_F(x_k)\| \\ & \quad + \rho \|J_F(x_k)\| \left(3 \times m^{\frac{2-q}{2}} \|F(x_{k+1}) - l_F(x_{k+1}; x_k)\|^{q-1} \right), \end{aligned}$$

where the second inequality follows from Hölder's inequality. Moreover, using the fact that for any vector $v \in \mathbb{R}^m$ and $q \in (1, 2]$, we have:

$$\|v\|_2 \leq \|v\|_q,$$